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Original Article

Safe Landing: Feasibility and Safety of Operating Room Extubation in Minimally Invasive Cardiac Valve Surgery

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Objective: This study was designed to explore key safety outcomes after operating room (OR) extubation in minimally invasive cardiac valve surgery.

Design: Single-center retrospective chart review.

Setting: Academic medical center in the United States.

Participants: Patients undergoing valvular surgery via thoracotomy (November 2017–October 2022) at a single institution.

Interventions: The OR extubation protocol was implemented on August 20, 2020.

Measurements and Main Results: Delirium rates, reintubation rates, and intubation duration were compared before and after OR extubation protocol implementation. Logistic regression identified patient perioperative characteristics associated with unsuccessful OR extubation.

Among 312 patients, 254 were extubated in the intensive care unit (ICU) and 58 in the OR. Preoperative demographics were comparable except for the Charlson Comorbidity Index (median: 2.0 ICU extubation v 1.5 OR extubation). Interrupted time series analysis showed no change in postoperative delirium post-OR extubation implementation, with a trend toward decreasing delirium (risk ratio = 0.37, CI: 0.13–1.10, $p = 0.07$). The postimplementation era also had a lower median intubation duration (8 hours v 13 hours, $p < 0.001$) without increasing reintubation rates (1.7% v 7.9%, $p = 0.159$). Increased bypass length (odds ratio = 0.99, CI: 0.98–0.99, $p < 0.001$), intraoperative morphine milligram equivalents (odds ratio = 0.99, CI: 0.99–1.0, $p = 0.009$), and preoperative Charlson Comorbidity Index above 3 (odds ratio = 0.42, CI: 0.19–0.95, $p = 0.037$) were associated with decreased odds of OR extubation.

Conclusions: OR extubation was not associated with increased postoperative delirium or reintubation rates but did decrease intubation duration. Successful OR extubation relies upon the consideration of various patient perioperative characteristics.

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Key Words: operating room extubation; successful operating room extubation; minimally invasive cardiac surgery; postoperative delirium; intubation duration; reintubation rates

DELIRIUM IS A COMMON postoperative complication occurring in up to 52% of cardiac surgery patients.¹ Post-cardiac surgery delirium has also been shown to be a

strong independent predictor of increased intensive care unit (ICU) length of stay and hospital charges,² as well as long term mortality up to 10 years.³ Several predictive factors are associated with postoperative delirium, including diabetes mellitus, cerebrovascular disease, peripheral vascular disease, and prolonged intubation,⁴ among which only intubation duration is potentially modifiable in the perioperative setting to

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reduce the risk of delirium. Prior studies in Canada⁵ and Europe^{6,7} have demonstrated an association between the length of intubation after operation and the risk of postoperative delirium. A prior study in Japan associated extubation in the operating room (OR) with lower postoperative delirium.⁸

However, OR extubation for cardiac cases is not widespread and only done for select patients. Given the historical reliance on opioid-heavy anesthetics for hemodynamic stability and heavy sedation required in cardiac surgery,⁹ it has been common practice to extubate patients in the ICU. This notion has been challenged recently, though differing conclusions have been reported. On the one hand, OR extubation protocols were safely implemented in isolated coronary artery bypass graft (CABG)¹⁰ and lung transplant cases.¹¹ Another study showed that younger age, lower body mass index, higher preoperative serum albumin, absence of chronic lung disease and diabetes, a less-invasive surgical approach, isolated coronary bypass surgery, elective surgery, and lower doses of intraoperative intravenous fentanyl were each independently associated with successful OR extubation after cardiac surgery, without an increased need for reintubation.¹² On the other hand, one study comparing OR extubation versus extubation within 6 hours of leaving the OR showed that OR extubation led to a greater need for reintubation and reoperation due to bleeding.¹³ Another study in Korea determined that OR extubation for cardiac surgery patients was associated with a higher incidence of postoperative hemodynamic instability and greater sedative requirements compared with ICU extubation.¹⁴ These conflicting data leave a gap in practice recommendations regarding OR extubation in cardiac surgery patients.

It remains unclear whether OR extubation is safe in cardiac patients, in whom this practice might be considered, and if it impacts a patient's postoperative course and their risk of postoperative delirium.

This study was designed to evaluate the feasibility of implementing an OR extubation protocol in a US hospital for minimally invasive valvular surgeries and investigate its effect on acute postoperative complications such as postoperative delirium and reintubation rates. The authors also aim to delineate what perioperative factors in this patient population portend successful OR extubation and hypothesize that if stable patients can be safely extubated in the OR instead of the ICU, the total intubation duration can be decreased without a concomitant increase in postoperative delirium and reintubation rates.

Methods

This study was approved by the Institutional Review Board at the authors' institution.

Patient Population and Study Design

All adults aged ≥ 18 who underwent cardiac valve repair or replacement via thoracotomy approach at the authors' institution during the study period from 11/2017 to 10/2022 were

retrospectively reviewed. Patients who had an isolated aortic valve repair or replacement were excluded.

Baseline patient demographics and perioperative characteristics were abstracted from the electronic health record. Patient preoperative psychiatric and opioid medication use was also captured. Each patient had a Charlson Comorbidity Index (CCI) calculated as a measure of preoperative baseline health status.¹⁵ Patients who tested positive for delirium on the confusion assessment method instrument or with the terms "delirium" or "altered mental status" without a clear or diagnosed cause upon chart review were considered delirious.

The primary aim was to determine the association between extubation location and postoperative delirium rates, with the primary outcome being postoperative delirium and secondary outcomes being reintubation rates and total intubation time. The secondary aim was to determine the association between successful OR extubation and perioperative patient characteristics.

Basic demographics, intraoperative variables, and postoperative variables were compared between patients extubated in the ICU versus patients extubated in the OR using Wilcoxon rank-sum tests and Chi-squared tests for continuous and categorical variables, respectively.

All analyses were conducted in R version 4.2.2 with an alpha level of 0.05.¹⁶ All confidence intervals were calculated at 95%.

Interrupted Time Series Analysis

An interrupted time series (ITS) analysis is used to understand how and if the incidence of a certain outcome has changed after an intervention was implemented at a specific time point. An OR extubation protocol was initiated in the summer of 2020, with the first occurrence of OR extubation on 08/20/2020, which was designated as the starting date of OR extubation implementation. An ITS with quasi-Poisson regression was used to analyze quarterly rate data on delirium.

Univariate Analysis

Patients were divided into an early era (11/01/2017-08/19/2020) and a late era (08/20/2020-10/31/2022) based on when the OR extubation protocol was initiated. A univariate analysis was performed to compare intubation duration between the two eras using the Wilcoxon rank-sum test.

Logistic Regression Analysis

Logistic regression was used to determine the association between postoperative delirium and extubation location, adjusting for bypass length, intraoperative morphine milligram equivalents (MMEs), and age. Logistic regression was also used on the binary outcome of OR extubation to determine its associations with patient age, prior depression and/or anxiety, CCI, intraoperative cardiopulmonary bypass length, and intraoperative MMEs.

Results

Three hundred and twelve patients within the study period met inclusion criteria, with 254 patients extubated in the ICU and 58 in the OR. The median age was 65 years in the ICU extubation group and 62 in the OR extubation group ($p = 0.168$), sex distribution was 55.5% female in the ICU extubation group and 55.2% female in the OR extubation group ($p = 0.999$), and median BMI was 26.0 in the ICU extubation group and 26.9 in the OR extubation group ($p = 0.347$). The OR extubation group had a lower median CCI score of 1.5 compared with 2.0 in the ICU extubation group ($p = 0.042$). Full preoperative patient demographics can be found in [Table 1](#).

Rates of mitral valve ($p = 0.973$) and tricuspid valve procedures ($p = 0.313$) were similar between the groups. Those extubated in the OR had lower operation (median: 274 minutes ν 312 minutes, $p < 0.001$) and bypass lengths (median 90 minutes ν 138 minutes, $p < 0.001$), higher rates of cross-clamp use (62.1% ν 26.4%, $p < 0.001$), and lower intraoperative MMEs (median: 125 ν 150, $p = 0.045$) compared with the ICU extubation group. Left ventricular ejection fraction both before ($p = 0.988$) and after initiation of cardiopulmonary bypass ($p = 0.250$) were comparable between study groups. Full intraoperative variables separated by extubation location can be found in [Table 2](#). Postoperative delirium (OR extubation 6.9% ν ICU extubation 13.4%, $p = 0.254$) and reintubation rates (OR extubation 1.7% ν ICU extubation 7.9%, $p = 0.159$) were comparable between the groups ([Table 3](#)). The incidence of OR extubation after protocol implementation is shown in [Supplementary Figure 1](#).

ITS analysis for quarterly delirium rates demonstrated comparable delirium rates before and after OR extubation, with a decreasing trend seen after the implementation of OR extubation (risk ratio [RR] = 0.37, confidence interval [CI]: 0.13–1.10, $p = 0.07$) ([Fig 1](#)). Older patients are more likely to experience postoperative delirium (odds ratio = 1.04, CI: 1.01–1.97, $p = 0.007$), while the odds of delirium are comparable for OR extubation and ICU extubation (odds ratio = 0.61, CI: 0.20–1.87, $p = 0.38$). Neither bypass length nor intraoperative MMEs were associated with postoperative delirium ([Fig 2, A](#)). A univariate analysis compared intubation duration before and after OR extubation began. In this analysis, the median and interquartile range [IQR] of intubation duration in hours was 13.0 [IQR: 9.59–18.34] in the early era and 7.76 hours [IQR: 4.29–12.15] in the late era ($p < 0.001$) ([Fig 3](#)).

Logistic regression for factors associated with OR extubation showed that increased bypass length (odds ratio = 0.99, CI: 0.98–0.99, $p < 0.001$), increased intraoperative MMEs (odds ratio = 0.99, CI: 0.99–1.0, $p = 0.009$), and preoperative CCI above 3 (odds ratio = 0.42, CI: 0.19–0.95, $p = 0.037$) all significantly decrease the odds of OR extubation ([Fig 2, B](#)). Age, history of anxiety and/or depression, and a mild-moderate CCI of 1–2 were not associated with extubation in the OR. The current guidelines for OR extubation in minimally invasive mitral valve surgery can be found in [Supplementary Figure 2](#).

Discussion

While OR extubation is becoming more common for certain cardiac cases such as isolated CABG,¹⁰ the safety of OR extubation has not been broadly established for additional cardiac surgical procedures. This study examined the safety and outcomes of OR extubation for valvular surgeries via thoracotomy. With post-cardiac surgery delirium linked to intubation practices,⁴ this study showed that initiating OR extubation does not lead to increased postoperative delirium rates ($p = 0.07$). Furthermore, [Figure 1](#) illustrates a potential downward trend in delirium rates after OR extubation initiation that may guide future studies and implementation of OR extubation in minimally invasive cardiac surgery. It is also possible that OR extubation can be combined with education and environmental modifications, including delirium management training for ICU personnel and light adjustments, which may further reduce delirium rates.¹⁷ Such a multidisciplinary set of interventions may lead to significant reductions in delirium rates.

The need for reintubation is a potential complication of premature extubation. Hawkins and colleagues compared patients extubated in the OR versus those extubated within 6 hours of leaving the OR for patients undergoing nonemergent CABG and/or valve operations and found that those undergoing OR extubation had higher reintubation rates (4.3% ν 1.8%, $p = 0.008$).¹³ However, the current study showed statistically comparable reintubation rates for OR versus ICU extubation (1.7% ν 7.9%, $p = 0.159$), indicating that, with appropriate patient selection, patients can safely be extubated in the OR without significantly increasing their reintubation rate.

With the results indicating no increase in postoperative delirium or reintubation rates after OR extubation, the significant decrease in intubation duration is notable given that longer intubation durations have been consistently associated with increased risk of delirium.^{4–7} Many studies have also linked early extubation in cardiac patients with decreased ICU and hospital lengths of stay,^{18–21} as well as other complications such as acute kidney injury²² and early postoperative weight gain as a marker for inflammation.²¹ Specifically, intubation durations above a 12-hour threshold have been shown to increase complications.^{5,7} The current study showed a dramatic drop in median intubation duration from 13 hours to less than 8 hours, suggesting the possibility that a higher powered study could associate OR extubation with improved patient outcomes and decreased delirium rates in this patient population.

The safety of OR extubation, in part, stems from the careful selection of patients for these protocols. Currently, the decision to extubate in the OR is based on preoperative patient characteristics, including age <65 years, body mass index <35 kg/m², lack of severe lung disease, and clinician judgment during the operation ([Supplementary Fig 1](#)). The current study design only included minimally invasive valvular surgeries, in accordance with a previous study showing that less invasive surgical approaches portend successful OR extubation.¹² The analysis also showed that shorter bypass length, decreased intraoperative opioid usage, and fewer

Table 1
Baseline Demographic and Comorbidity Variables Grouped by Patients Extubated in the ICU or OR

	Extubated in ICU (N = 254)	Extubated in OR (N = 58)	Overall (N = 312)	p-Value [*]
Age				
Mean (SD)	61.8 (15.9)	57.9 (17.7)	61.1 (16.3)	0.168
Median [Q1, Q3]	64.5 [53.0, 74.0]	61.5 [45.2, 72.8]	64.0 [51.0, 74.0]	
Sex				
Female	141 (55.5%)	32 (55.2%)	173 (55.4%)	1
Male	113 (44.5%)	26 (44.8%)	139 (44.6%)	
Race				
White	151 (59.4%)	41 (70.7%)	192 (61.5%)	0.469
Black	16 (6.3%)	2 (3.4%)	18 (5.8%)	
Missing	87 (34.3%)	15 (25.9%)	102 (32.7%)	
Ethnicity				
Not Hispanic/Latino	26 (10.2%)	10 (17.2%)	36 (11.5%)	0.965
Hispanic/Latino	2 (0.8%)	0 (0%)	2 (0.6%)	
Missing	226 (89.0%)	48 (82.8%)	274 (87.8%)	
BMI				
Mean (SD)	27.3 (6.2)	27.3 (5.1)	27.3 (6.0)	0.532
Median [Q1, Q3]	26.3 [22.9, 30.3]	26.5 [23.4, 31.3]	26.4 [22.9, 30.4]	
History of diabetes mellitus				
No	197 (77.6%)	47 (81.0%)	244 (78.2%)	0.799
Yes: uncomplicated	49 (19.3%)	9 (15.5%)	58 (18.6%)	
Yes: end-organ damage	8 (3.1%)	2 (3.4%)	10 (3.2%)	
History of hypertension				
No	100 (39.4%)	23 (39.7%)	123 (39.4%)	1
Yes	154 (60.6%)	35 (60.3%)	189 (60.6%)	
History of CVA or cerebrovascular disease				
No	217 (85.4%)	47 (81.0%)	264 (84.6%)	0.525
Yes	37 (14.6%)	11 (19.0%)	48 (15.4%)	
History of dementia				
No	253 (99.6%)	58 (100%)	311 (99.7%)	1
Yes	1 (0.4%)	0 (0%)	1 (0.3%)	
History of anxiety/depression				
No	177 (69.7%)	44 (75.9%)	221 (70.8%)	0.463
Yes	76 (29.9%)	14 (24.1%)	90 (28.8%)	
Missing	1 (0.4%)	0 (0%)	1 (0.3%)	
History of chronic pain				
No	232 (91.3%)	54 (93.1%)	286 (91.7%)	0.861
Yes	22 (8.7%)	4 (6.9%)	26 (8.3%)	
Use of preoperative dementia meds				
No	251 (98.8%)	58 (100%)	309 (99.0%)	0.931
Yes	3 (1.2%)	0 (0%)	3 (1.0%)	
Use of preoperative SSRIs or benzodiazepines				
No	199 (78.3%)	44 (75.9%)	243 (77.9%)	0.813
Yes	55 (21.7%)	14 (24.1%)	69 (22.1%)	
Updated Charlson Comorbidity Index				
Mean (SD)	2.1 (2.1)	1.5 (1.6)	2.0 (2.0)	0.042
Median [Q1, Q3]	2.0 [0.0, 3.0]	1.5 [0.0, 2.0]	2.0 [0.0, 3.0]	
Updated Charlson Comorbidity Index				
0	82 (32.3%)	26 (44.8%)	108 (34.6%)	NA
1	10 (3.9%)	3 (5.2%)	13 (4.2%)	
2	75 (29.5%)	16 (27.6%)	91 (29.2%)	
3	38 (15.0%)	4 (6.9%)	42 (13.5%)	
4	30 (11.8%)	8 (13.8%)	38 (12.2%)	
5	5 (2.0%)	0 (0%)	5 (1.6%)	
6	4 (1.6%)	0 (0%)	4 (1.3%)	
7	2 (0.8%)	1 (1.7%)	3 (1.0%)	
8	2 (0.8%)	0 (0%)	2 (0.6%)	
9	4 (1.6%)	0 (0%)	4 (1.3%)	
10	1 (0.4%)	0 (0%)	1 (0.3%)	
11	0 (0%)	0 (0%)	0 (0%)	
12	1 (0.4%)	0 (0%)	1 (0.3%)	

(continued)

Table 1 (continued)

	Extubated in ICU (N = 254)	Extubated in OR (N = 58)	Overall (N = 312)	p-Value [*]
Updated Charlson Comorbidity Index				
None (0)	82 (32.3%)	26 (44.8%)	108 (34.6%)	0.121
Mild/moderate (1-2)	85 (33.5%)	19 (32.8%)	104 (33.3%)	
Severe (3+)	87 (34.3%)	13 (22.4%)	100 (32.1%)	

Abbreviations: BMI, body mass index; CVA, cerebrovascular accident; ICU, intensive care unit; OR, operating room; SD, standard deviation; SSRI, selective serotonin reuptake inhibitor.

^{*} These are nominal p-values and are not adjusted for multiple comparisons. Wilcoxon rank-sum and Chi-squared tests are used for continuous and categorical variables, respectively.

Table 2

Intraoperative Variables Grouped by Patients Extubated in the ICU or OR

	Extubated in ICU (N = 254)	Extubated in OR (N = 58)	Overall (N = 312)	p-Value [*]
Mitral valve procedure				
No	45 (17.7%)	11 (19.0%)	56 (17.9%)	0.973
Yes	209 (82.3%)	47 (81.0%)	256 (82.1%)	
Tricuspid valve procedure				
No	187 (73.6%)	47 (81.0%)	234 (75.0%)	0.313
Yes	67 (26.4%)	11 (19.0%)	78 (25.0%)	
Prebypass LVEF				
Mean (SD)	53.0 (6.2)	53.5 (4.3)	53.1 (5.9)	0.988
Median [Q1, Q3]	55.0 [55.0, 55.0]	55.0 [55.0, 55.0]	55.0 [55.0, 55.0]	
Missing	4 (1.6%)	1 (1.7%)	5 (1.6%)	
Postbypass LVEF				
Mean (SD)	52.3 (6.7)	53.7 (4.1)	52.6 (6.4)	0.250
Median [Q1, Q3]	55.0 [55.0, 55.0]	55.0 [55.0, 55.0]	55.0 [55.0, 55.0]	
Missing	4 (1.6%)	2 (3.4%)	6 (1.9%)	
Operation length (min)				
Mean (SD)	339.9 (101.0)	293.9 (67.1)	331.4 (97.2)	<0.001
Median [Q1, Q3]	315.5 [274.0, 384.0]	274.0 [252.0, 342.8]	308.0 [264.5, 376.0]	
Bypass length (min)				
Mean (SD)	153.4 (77.4)	107.5 (43.2)	144.7 (74.3)	<0.001
Median [Q1, Q3]	138.0 [93.0, 188.0]	90.0 [76.2, 131.8]	126.5 [87.2, 182.0]	
Missing	6 (2.4%)	0 (0%)	6 (1.9%)	
Cross-clamp used during surgery				
No	187 (73.6%)	22 (37.9%)	209 (67.0%)	<0.001
Yes	67 (26.4%)	36 (62.1%)	103 (33.0%)	
Erector spinae block				
No	132 (52.0%)	2 (3.4%)	134 (42.9%)	<0.001
Yes	122 (48.0%)	56 (96.6%)	178 (57.1%)	
Lowest hematocrit recorded during surgery				
Mean (SD)	26.8 (5.0)	28.6 (4.8)	27.1 (5.0)	0.016
Median [Q1, Q3]	26.0 [23.0, 30.0]	28.0 [25.5, 32.0]	26.0 [23.0, 30.8]	
Missing	3 (1.2%)	3 (5.2%)	6 (1.9%)	
Estimated blood loss during surgery				
Mean (SD)	581.5 (462.3)	554.9 (519.3)	576.5 (472.6)	0.22
Median [Q1, Q3]	500.0 [300.0, 787.5]	500.0 [200.0, 600.0]	500.0 [300.0, 750.0]	
Intraoperative intravenous opioids				
No	3 (1.2%)	0 (0%)	3 (1.0%)	0.931
Yes	251 (98.8%)	58 (100%)	309 (99.0%)	
Intraoperative MMEs				
Mean (SD)	169.4 (113.5)	130.0 (63.0)	162.1 (107.0)	0.045
Median [Q1, Q3]	150.0 [75.0, 225.0]	125.0 [80.0, 167.5]	150.0 [75.0, 225.0]	

Abbreviations: ICU, intensive care unit; LVEF, left ventricular ejection fraction; MMEs, morphine milligram equivalents; OR, operating room; SD, standard deviation.

^{*} These are nominal p-values and are not adjusted for multiple comparisons. Wilcoxon rank-sum and Chi-squared tests are used for continuous and categorical variables, respectively.

Table 3
Postoperative Delirium and Need for Reintubation Grouped by Patients Extubated in the ICU or OR

	Extubated in ICU (N = 254)	Extubated in OR (N = 58)	Overall (N = 312)	p-Value*
Postoperative delirium				
No	220 (86.6%)	54 (93.1%)	274 (87.8%)	0.254
Yes	34 (13.4%)	4 (6.9%)	38 (12.2%)	
Reintubation				
No	232 (91.3%)	57 (98.3%)	289 (92.6%)	0.159
Yes	20 (7.9%)	1 (1.7%)	21 (6.7%)	
Missing	2 (0.8%)	0 (0%)	2 (0.6%)	

Abbreviations: ICU, intensive care unit; OR, operating room.

*These are nominal p-values and are not adjusted for multiple comparisons. Wilcoxon rank-sum and Chi-squared tests are used for continuous and categorical variables, respectively.

comorbidities as measured by the CCI, are significantly associated with OR extubation for these patients. These findings could be a starting point for careful patient identification in a protocolized approach to OR extubation in select cardiac procedures. A guideline serving such a purpose would likely incorporate additional preoperative patient characteristics and intraoperative variables that measure the complexity of the procedure, such as bypass length and intraoperative opioid dosage. Along with the data supporting the safety of OR extubation in a subpopulation of cardiac surgery patients, it is possible that OR extubation could be safely implemented for some selected, lower-risk cardiac patients undergoing straightforward operations.

In summary, the current results demonstrate that initiation of an OR extubation protocol for minimally invasive valvular surgery is noninferior to the standard of care, which is ICU extubation when comparing delirium and reintubation rates, while allowing for significantly decreased total intubation time. Though this study was designed to show the

noninferiority of OR extubation compared with ICU extubation, the observed decrease in intubation duration may serve as a launchpad for larger studies to investigate whether OR extubation leads to superior outcomes. Furthermore, the specific finding that select patients can be safely extubated sooner in the OR rather than waiting to be extubated in the ICU may also improve clinical workflows and hospital efficiency. Lastly, this work provides a foundation to create a protocolized approach for determining who can be safely extubated in the OR so that OR extubation may be implemented more widely in cardiac surgery.

Limitations and Future Directions

As a single-center, retrospective study, this analysis is subject to the selection biases and influence of the authors' institutional practice. Since this study focused on valvular surgeries via thoracotomy, future studies are needed to investigate whether these findings are generalizable to other cardiac

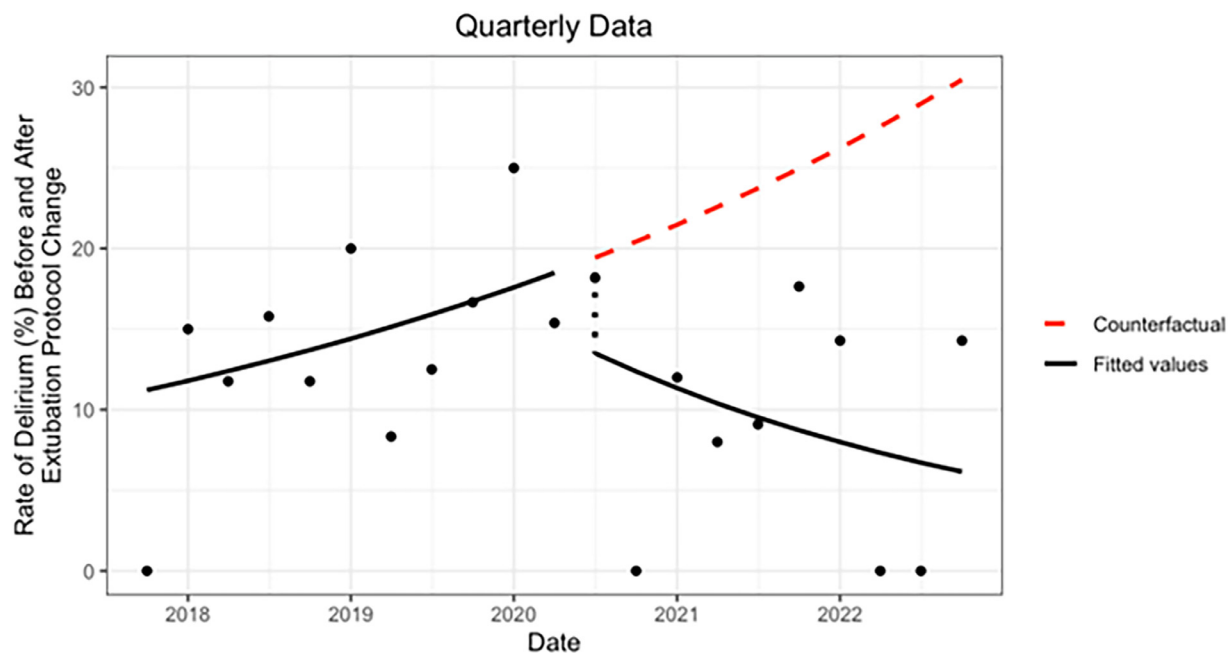


Fig 1. Interrupted time series analysis of delirium rate before and initiation of OR extubation protocol. There is a decreased rate of delirium after OR extubation began (first occurrence: August 20, 2020) with the actual delirium rate (solid black line) compared with the counterfactual (red dotted line). (relative risk = 0.37, 2.5% CI: 0.13, 97.5% CI: 1.10, $p = 0.07$). OR, operating room.

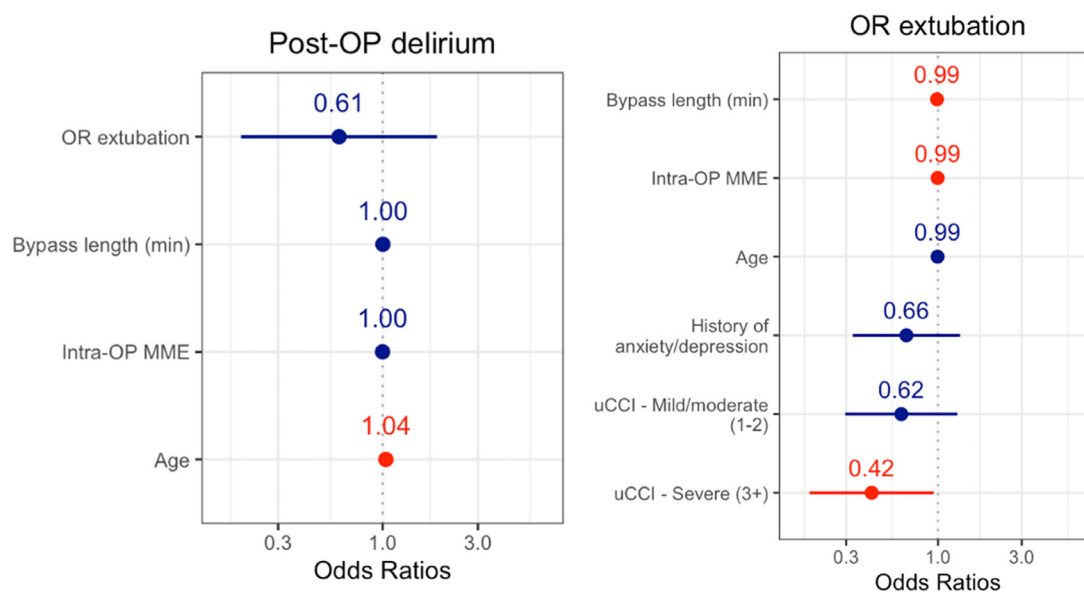


Fig 2. Forest plots of odds ratios for covariates in the logistic regression models. (Left) Logistic regression model with postoperative delirium as the outcome, with covariates of OR extubation, bypass length, intraoperative MMEs and patient age. Increased age is significantly associated with increased postoperative delirium ($p = 0.007$). (Right) Logistic regression model with OR extubation as the outcome, with bypass length, intraoperative MMEs, patient age, patient history of depression and/or anxiety, preoperative mild-moderate CCI, and preoperative severe CCI as predictor variables. Increased bypass length ($p < 0.001$), intraoperative MMEs ($p = 0.009$), and severe CCI ($p = 0.037$) are associated with decreased odds of OR extubation. CCI, Charlson Comorbidity Index; MMEs, morphine milligram equivalents; OR, operating room.

surgeries, including those requiring a sternotomy. Confounders such as different providers, increased use of multimodal anesthetics over opioids during the study timeframe, and a modern emphasis on delirium prevention could also obscure the true effect or lack thereof of OR extubation on postoperative delirium. Specifically, the authors' institution began widespread use of erector spinae blocks in January of 2019, potentially contributing to the decreased intubation times and downtrend in delirium seen in this new era. Another notable practice shift that could be a confounder is that the authors' main surgeon, used to fibrillate for mitral valve replacements but now almost exclusively uses a cross-clamp, which likely explains the significant difference in cross-clamp rates between those extubated in the ICU versus OR despite similar rates of mitral and tricuspid operations.

As a feasibility study, this study was focused primarily on acute postoperative complications, namely the need for reintubation and the development of delirium. Future studies could elucidate how OR extubation affects each patient's

clinical course (with hemodynamic, metabolic, and transfusion data) and lengths of stay. Higher-powered studies or propensity-matched cohorts are needed to explore significant differences in outcomes to help elucidate the range of benefits and risks that OR extubation has on the patient's full clinical course. These outcomes could include delirium rates, ventilator-associated outcomes such as hospital-acquired pneumonia, and the need for noninvasive ventilation, reoperation, and acute kidney injury. Along with a full accounting of the risks and benefits, a cost analysis of the practice of OR extubation may provide additional context for how it might fit into an overall patient care plan.

Conclusions

This study showed that initiation of OR extubation for patients undergoing valvular surgery via thoracotomy is not associated with increased postoperative delirium or reintubation rates compared with the standard of ICU extubation, while significantly decreasing intubation duration. Key perioperative variables that portend successful OR extubation were also identified. Therefore, OR extubation could be a promising method for decreasing intubation duration without increasing postoperative complications for minimally invasive cardiac surgery patients.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

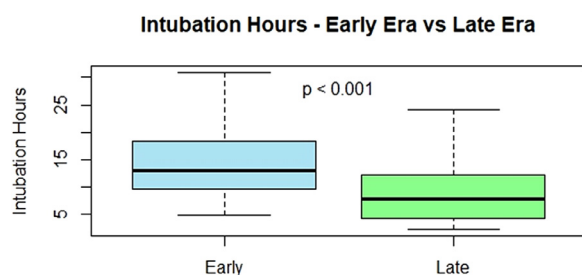


Fig 3. Boxplot comparing intubation duration in the early and late eras. There was a significant decrease in total intubation duration in the late era after OR extubation was initiated compared with the early era before OR extubation (median: 7.76 h v 13 h, $p < 0.001$). OR, operating room.

CRediT authorship contribution statement

Chen Chia Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation, Conceptualization. **Alexandra DeBose-Scarlett:** Writing – review & editing, Methodology, Investigation, Data curation, Conceptualization. **Rebecca Irlmeier:** Methodology, Investigation, Formal analysis. **Fei Ye:** Methodology, Investigation, Formal analysis. **Kara Siegrist:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Ashish S. Shah:** Writing – review & editing, Supervision, Project administration, Conceptualization. **Meredith Kingeter:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Investigation, Conceptualization.

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Supplementary materials

Supplementary material associated with this article can be found in the online version at [doi:10.1053/j.jvca.2024.09.014](https://doi.org/10.1053/j.jvca.2024.09.014).

References

- Rudolph JL, Jones RN, Levkoff SE, et al. Derivation and validation of a preoperative prediction rule for delirium after cardiac surgery. *Circulation* 2009;119:229–36. <https://doi.org/10.1161/CIRCULATIONAHA.108.795260>.
- Brown CH, Laflam A, Max L, et al. The impact of delirium after cardiac surgical procedures on postoperative resource use. *Ann Thorac Surg* 2016;101:1663–9. <https://doi.org/10.1016/j.athoracsur.2015.12.074>.
- Gottesman RF, Grega MA, Bailey MM, et al. Delirium after coronary artery bypass graft surgery and late mortality. *Ann Neurol* 2010;67:338–44. <https://doi.org/10.1002/ana.21899>.
- Nikolić BD, Putnik SM, Lazovic DM, et al. Can we identify risk factors for postoperative delirium in cardiac coronary patients? Our experience. *Heart Surg Forum* 2012;15:195. <https://doi.org/10.1532/HSF98.20111166>.
- Muller Moran HR, Maguire D, Maguire D, et al. Association of earlier extubation and postoperative delirium after coronary artery bypass grafting. *J Thorac Cardiovasc Surg* 2020;159. <https://doi.org/10.1016/j.jtcvs.2019.03.047>;182-90.e7.
- Kazmierski J, Kowman M, Banach M, et al. Incidence and predictors of delirium after cardiac surgery: Results from the IPDACS study. *J Psychosom Res* 2010;69:179–85. <https://doi.org/10.1016/j.jpsychores.2010.02.009>.
- Kotfis K, Szylińska A, Listewnik M, et al. Balancing intubation time with postoperative risk in cardiac surgery patients—A retrospective cohort analysis. *Ther Clin Risk Manag* 2018;14:2203–12. <https://doi.org/10.2147/TCRM.S182333>.
- Kobayashi T, Kurazumi H, Gohra H, et al. [Effect of ultra-fast-track extubation in cardiovascular surgery]. [Article in Japanese]. *Kyobu Geka* 2016;69:971–8.
- Grant MC, Chappell D, Gan TJ, et al. Pain management and opioid stewardship in adult cardiac surgery: Joint Consensus Report of the PeriOperative Quality Initiative and the Enhanced Recovery After Surgery Cardiac Society. *J Thorac Cardiovasc Surg* 2023;166:1695–706. <https://doi.org/10.1016/j.jtcvs.2023.01.020>;e2.
- James L, Smith DE, Galloway AC, et al. Routine extubation in the operating room after isolated coronary artery bypass. *Ann Thorac Surg* 2024;117:87–94. <https://doi.org/10.1016/j.athoracsur.2023.09.031>.
- Ai Q, Deng H, Huang D, et al. Early extubation in operating room after single-lung transplantation: A single institutional experience. *Ann Palliat Med* 2021;10:4134–42. <https://doi.org/10.21037/apm-20-1598>.
- Subramaniam K, DeAndrade DS, Mandell DR, et al. Predictors of operating room extubation in adult cardiac surgery. *J Thorac Cardiovasc Surg* 2017;154:1656–65. <https://doi.org/10.1016/j.jtcvs.2017.05.107>;e2.
- Hawkins AD, Strobel RJ, Mehaffey JH, et al. Operating room versus intensive care unit extubation within 6 hours after on-pump cardiac surgery: Early results and hospital costs. *Semin Thorac Cardiovasc Surg* 2022; S1043-0679:00272–6. <https://doi.org/10.1053/j.semtcvs.2022.09.013>.
- Kim CH, Lee JH, Kwon HW, et al. Extubation in operating room versus early extubation in ICU after open-heart surgery in patients with CHDs. *Cardiol Young* 2024;34:914–8. <https://doi.org/10.1017/S1047951123003839>.
- Coyan GN, Chin H, Shah A, et al. Charlson Comorbidity Index is associated with longer-term mortality and re-admissions following coronary artery bypass grafting. *J Surg Res* 2022;275:300–7. <https://doi.org/10.1016/j.jss.2022.02.012>.
- R Core Team. R: A language and environment for statistical computing. Vienna, Austria: R Foundation for Statistical Computing; 2021 <https://www.R-project.org/>.
- Tehranehsat B, Hosseinpour N, Mani A, et al. The effect of multi-component interventions on the incidence rate, severity, and duration of post open heart surgery delirium among hospitalized patients. *J Cardiothorac Surg* 2021;16:32. <https://doi.org/10.1186/s13019-021-01422-0>.
- Salhiyyah K, Elsobky S, Raja S, et al. A clinical and economic evaluation of fast-track recovery after cardiac surgery. *Heart Surg Forum* 2011;14:330. <https://doi.org/10.1532/HSF98.20111029>.
- Yanatori M, Tomita S, Miura Y, et al. Feasibility of the fast-track recovery program after cardiac surgery in Japan. *Gen Thorac Cardiovasc Surg* 2007;55:445–9. <https://doi.org/10.1007/s11748-007-0162-2>.
- Dunstan JL, Riddle MM. Rapid recovery management: The effects on the patient who has undergone heart surgery. *Heart Lung* 1997;26:289–98. [https://doi.org/10.1016/S0147-9563\(97\)90086-8](https://doi.org/10.1016/S0147-9563(97)90086-8).
- Engelman RM, Rousou JA, Flack JE, et al. Fast-track recovery of the coronary bypass patient. *Ann Thorac Surg* 1994;58:1742–6. [https://doi.org/10.1016/0003-4975\(94\)91674-8](https://doi.org/10.1016/0003-4975(94)91674-8).
- Heringlake M, Nowak Y, Schön J, et al. Postoperative intubation time is associated with acute kidney injury in cardiac surgical patients. *Crit Care* 2014;18:547. <https://doi.org/10.1186/s13054-014-0547-4>.